

Mark R. Saddler

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Education

Massachusetts Institute of Technology Cambridge, MA
Ph.D. in Computational Neuroscience 2017 – 2024
Thesis: “Task-optimized models of human hearing link perception and neural coding”

The University of Chicago Chicago, IL
B.A. in Biological Sciences (specialization in neuroscience) 2013 – 2017
B.S. in Biological Chemistry
B.A. in Chemistry

Research Positions

Technical University of Denmark Lyngby, Denmark
Postdoc, Department of Health Technology (advised by Torsten Dau) Feb 2024 – Present

Meta Reality Labs Redmond, WA
Research Scientist Intern, Audio (advised by Antje Ihlefeld) May 2022 – Sep 2022

Massachusetts Institute of Technology Cambridge, MA
Ph.D., Department of Brain and Cognitive Sciences (advised by Josh McDermott) Sep 2017 – Feb 2024

Woods Hole Oceanographic Institution Woods Hole, MA
Summer Student Fellow, Biology Department (advised by Laela Sayigh) Jun 2016 – Aug 2016

The University of Chicago Chicago, IL
Undergraduate Research Assistant, Epilepsy Lab (advised by Wim van Drongelen) Oct 2015 – Jun 2017

UCSD Scripps Institution of Oceanography La Jolla, CA
Research Intern, Marine Physical Laboratory (advised by Simone Baumann-Pickering) Jun 2015 – Sep 2015

Salk Institute for Biological Studies La Jolla, CA
Research Intern, Molecular Neurobiology Laboratory (advised by Sreekanth Chalasani) Jun 2014 – Sep 2014

Peer Reviewed Publications

- **MR Saddler**, JH McDermott (2024). Models optimized for real-world tasks reveal the task-dependent necessity of precise temporal coding in hearing. *Nature Communications*, 15, 10590. 
- **MR Saddler***, R Gonzalez*, JH McDermott (2021). Deep neural network models reveal interplay of peripheral coding and stimulus statistics in pitch perception. *Nature Communications*, 12, 7278. 
- **MR Saddler***, A Franci*, J Feather, K Qian, Y Zhang, JH McDermott (2021). Speech denoising with auditory models. *Proc. Interspeech*, 2021, 2681-2685. 
- **MR Saddler**, A Bocconcelli, LS Hickmott, G Chiang, R Landea-Briones, PA Bahamonde, G Howes, PS Segre, LS Sayigh (2017). Characterizing Chilean blue whale vocalizations with DTAGs: a test of using tag accelerometers for caller identification. *Journal of Experimental Biology*, 220, 4119-4129. 

Preprints and Proceedings

- **MR Saddler**, T Dau, JH McDermott (2025). Towards individualized models of hearing-impaired speech perception. *Proc. The 6th Clarity Workshop on Improving Speech-in-Noise for Hearing Devices*, 7-11. 
- A Banerjee, **MR Saddler**, JG Arenberg, JH McDermott (2025). A deep learning framework for understanding cochlear implants. *bioRxiv*. 
- **MR Saddler**, T Dau, JH McDermott (2024). Psychoacoustic phenomena explained by auditory task optimization. *Proc. Cognitive Computational Neuroscience*. 
- A Banerjee, **MR Saddler**, JH McDermott (2024). Neural network models of hearing clarify factors limiting cochlear implant outcomes. *Proc. Cognitive Computational Neuroscience*. 
- **MR Saddler**, M Khaleghi-Meybodi, A Ihlefeld (2023). Automated audio realism detection. *Technical Disclosure Commons*. 
- J Suresh, **MR Saddler**, V Bindokas, A Bhansali, L Pesce, J Wang, J Marks, W van Drongelen (2023). Emerging activity patterns and synaptogenesis in dissociated hippocampal cultures. *bioRxiv*. 
- A Bocconcelli, L Hickmott, G Chiang, P Bahamonde, G Howes, R Landea-Briones, F Caruso, **MR Saddler**, L Sayigh (2016). DTAG studies of blue whales (*Balaenoptera musculus*) in the Gulf of Corcovado, Chile. *Proc. Meetings on Acoustics* 27, 040002. 

Funding and Awards

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| • William Demant Foundation postdoctoral research grant (2.9 million DKK $\approx$ \$450,000 over 3 years) | 2025 |
| • Travel Award (Association for Research in Otolaryngology) | 2023 |
| • Best Video Pitch Award (Virtual Conference on Computational Audiology) | 2022 |
| • Travel Award (Association for Research in Otolaryngology) | 2022 |
| • K. Lisa Yang Integrative Computational Neuroscience Center Graduate Student Fellowship (MIT) | 2021 |
| • Travel Grant (Interspeech) | 2021 |
| • Best Poster Award (Advances and Perspectives in Auditory Neuroscience) | 2020 |
| • Angus MacDonald Award for Excellence in Undergraduate Teaching (MIT) | 2020 |
| • Henry E. Singleton Graduate Student Fellowship (MIT) | 2018 |
| • Best Poster Award (IBM AI Week) | 2019 |
| • Travel Award (Association for Research in Otolaryngology) | 2019 |
| • Best Student Paper Award (Acoustical Society of America) | 2017 |
| • Phi Beta Kappa (UChicago) | 2017 |
| • Francis E. Knock Prize in Biological Chemistry for highest GPA in major (UChicago) | 2017 |
| • Conference Travel Grant (Woods Hole Oceanographic Institution) | 2016 |
| • Summer Student Fellowship (Woods Hole Oceanographic Institution) | 2016 |
| • University Scholar Award (UChicago) | 2013 |
| • National Merit Scholarship | 2013 |
| • National AP Scholar | 2013 |

Invited Talks

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| • Danish Acoustic Society. Lyngby, Denmark. | Oct 2026 |
| • Forum Acusticum (session on computational and model-based approaches). Graz, Austria. | Sep 2026 |
| • Virtual Conference on Computational Audiology (pre-conference tutorial on machine learning). | Jun 2026 |
| • Association for Research in Otolaryngology (symposium on pitch). San Juan, Puerto Rico. | Feb 2026 |
| • Foundation Pour L'Audition, Auditory Horizons Workshop on AI. Paris, France. | Nov 2025 |
| • Pasteur Institut de L'Audition. Paris, France. | Nov 2024 |
| • DTU, Hearing Systems Presentation Day. Lyngby, Denmark. | Oct 2024 |
| • WSAudiology. Lyngby, Denmark. | Oct 2024 |
| • Universität Oldenburg, Department of Medical and Physical Acoustics. Oldenburg, Germany. | Jul 2024 |

- DTU, Hearing Systems Research Seminar. Lyngby, Denmark. Jun 2023
- Universität Graz, Research Seminar in Systematic Musicology. Graz, Austria. Jun 2023
- Ecole Normale Supérieure, Laboratoire des Systèmes Perceptifs. Paris, France. Apr 2022
- MIT, Department of Brain and Cognitive Sciences, Cog Lunch. Cambridge, MA. Apr 2022

Teaching

DTU MSc Programme in Engineering Acoustics

- Guest lecturer on machine learning: Auditory Signal Processing and Perception (22003) Spring 2026
- Guest lecturer on machine learning: Auditory Signal Processing and Perception (22003) Spring 2025

MIT Department of Brain and Cognitive Sciences

- TA: Machine-Motivated Human Vision (9.60) Spring 2023
- TA: Machine-Motivated Human Vision (9.60) Spring 2022
- TA: Introduction to Neural Computation (9.40) Spring 2020
- TA: Perception (9.35) Spring 2019

MIT Educational Studies Program

- HSSP Teacher: Sensational Neuroscience Summer 2019
- Splash! Instructor: Braaaaaains! The Neuroscience of Zombies Fall 2018
- HSSP Teacher: The Foundations of Intelligence: An Intro to Neuroscience Summer 2018
- HSSP Teacher: Memory and Mind Control: An Intro to Neuroscience Spring 2018
- Splash! Instructor: Introduction to Neuroscience Fall 2017

UChicago Biological Sciences Collegiate Division

- Lab TA: Biological Dynamics (BIOS 20236) Spring 2017
- Lab TA: Principles of Physiology (BIOS 20242) Fall 2016
- Lecture TA: Principles of Physiology (BIOS 20242) Fall 2015

Service

Peer Reviewing

Journal of the Acoustical Society of America; Nature Communications; Nature Machine Intelligence; eLife; PLoS Computational Biology; Attention, Perception, & Psychophysics; IEEE Transactions on Audio, Speech and Language Processing; Audio Engineering Society; Trends in Hearing

Meeting Organizing

- Forum Acusticum: co-organizer for “Computational and AI Approaches in Audiology” Sep 2026
- ISAAR: session chair for “Computational Audiology, AI, and Auditory Profiling” Aug 2025

Research Supervising

- DTU Engineering Acoustics MSc Thesis Supervisor 2026
- MIT Center for Brains Minds and Machines Summer Research Program Mentor 2020, 2021

Undergraduate Mentoring

- Graduate Resident Advisor in Baker House at MIT Aug 2018 – Jul 2023
- Assistant Resident Head (RA) of Thangaraj / Tufts House at UChicago Apr 2014 – Jun 2017

MIT OpenMind High-Performance Computing Cluster, Group Representative Jun 2019 – Feb 2024

2026

- **MR Saddler**, T Dau, JH McDermott. Perceptual sensitivity explained by optimization for ecological hearing tasks. *Forum Acusticum*. (invited talk)
- **MR Saddler**, T Dau, JH McDermott. Perceptual sensitivity is explained by optimization for ecological hearing tasks. *Virtual Conference on Computational Audiology*. (featured talk)
- **MR Saddler**. A practical introduction to machine learning for hearing science. *Virtual Conference on Computational Audiology*. (invited tutorial)
- A Banerjee, **MR Saddler**, JG Arenberg, JH McDermott. A deep learning framework for screening and developing new cochlear implants. *Association for Research in Otolaryngology*. (poster)
- **MR Saddler**, T Dau, JH McDermott. Hearing-impaired deep neural networks predict real-world hearing difficulties. *Association for Research in Otolaryngology*. (poster)
- **MR Saddler**. Understanding pitch perception with deep learning. *Association for Research in Otolaryngology*. (invited talk)

2025

- **MR Saddler**, T Dau, JH McDermott. Towards individualized models of hearing-impaired speech perception. *The 6th Clarity Workshop on Improving Speech-in-Noise for Hearing Devices*. (talk)
- **MR Saddler**, T Dau, JH McDermott. Modeling normal and impaired hearing with deep neural networks optimized for ecological tasks. *International Symposium on Auditory and Audiological Research*. (poster)
- **MR Saddler**, T Dau, JH McDermott. Modeling normal and impaired hearing with deep neural networks optimized for ecological tasks. *International Symposium on Hearing*. (talk)
- **MR Saddler**, T Dau, JH McDermott. Modeling normal and impaired hearing with deep neural networks optimized for ecological tasks. *Joint Annual Meeting of the German and Danish Acoustical Societies*. (talk)
- A Banerjee, **MR Saddler**, JG Arenberg, JH McDermott. Understanding cochlear implants using machine learning. *Association for Research in Otolaryngology*. (poster)
- A Magaro, EN Shook, AJE Kell, **MR Saddler**, G Elbanna, JH McDermott. Optimization under ecological realism reproduces signatures of human speech perception. *Association for Research in Otolaryngology*. (poster)
- **MR Saddler**, T Dau, JH McDermott. Modeling normal and impaired hearing with deep neural networks optimized for ecological tasks. *Association for Research in Otolaryngology*. (poster blitz)

2024

- A Banerjee, **MR Saddler**, JH McDermott. Neural network models of hearing clarify factors limiting cochlear implant outcomes. *Conference on Cognitive Computational Neuroscience*. (poster)
- **MR Saddler**, T Dau, JH McDermott. Psychoacoustic phenomena explained by auditory task optimization. *Conference on Cognitive Computational Neuroscience*. (poster)
- A Banerjee, **MR Saddler**, JH McDermott. Neural network models of hearing clarify factors limiting cochlear implant outcomes. *Virtual Conference on Computational Audiology*. (talk)
- **MR Saddler**, T Dau, JH McDermott. Modeling normal and impaired hearing with artificial neural networks optimized for ecological tasks. *Virtual Conference on Computational Audiology*. (talk)
- A Banerjee, **MR Saddler**, JH McDermott. Understanding cochlear implants using machine learning. *Association for Research in Otolaryngology*. (poster)
- A Magaro, EN Shook, AJE Kell, **MR Saddler**, JH McDermott. Optimization under ecological realism reproduces signatures of human speech perception. *Association for Research in Otolaryngology*. (poster)
- **MR Saddler**, JH McDermott. The role of temporal coding in hearing: evidence from task optimization. *Association for Research in Otolaryngology*. (poster)

2023

- **MR Saddler**, JH McDermott. The role of temporal coding in real-world hearing: evidence from machine learning. *International Symposium on Auditory and Audiological Research*. (talk)

- A Banerjee, **MR Saddler**, JH McDermott. Neural network models clarify the role of plasticity in cochlear implant outcomes. *Conference on Implantable Auditory Prostheses*. (poster)
- A Banerjee, **MR Saddler**, JH McDermott. Neural network models of hearing through a cochlear implant. *Association for Research in Otolaryngology*. (poster)
- IM Griffith*, MJ McPherson*, **MR Saddler***, JH McDermott (February, 2023). Task-optimized models of relative pitch. *Association for Research in Otolaryngology*. (poster)
- **MR Saddler**, JH McDermott. The role of temporal coding in real-world hearing: evidence from machine learning. *Association for Research in Otolaryngology*. (poster)

2022

- **MR Saddler**, JH McDermott. The role of temporal coding in real-world hearing: evidence from machine learning. *Advances and Perspectives in Auditory Neuroscience*. (poster)
- **MR Saddler**, JH McDermott. The role of temporal coding in everyday hearing: evidence from machine learning. *Virtual Conference on Computational Audiology*. (talk)
- **MR Saddler**, JH McDermott. The role of temporal coding in everyday hearing: evidence from deep neural networks. *COSYNE: Computational and Systems Neuroscience*. (poster)
- **MR Saddler**, JH McDermott. The role of auditory nerve phase locking in audition: evidence from deep neural networks. *Association for Research in Otolaryngology*. (poster)

2021

- **MR Saddler**, JH McDermott. The role of auditory nerve phase-locking in human hearing: evidence from deep neural networks. *Advances and Perspectives in Auditory Neuroscience*. (talk)
- **MR Saddler***, A Franci*, J Feather, K Qian, Y Zhang, JH McDermott. Speech denoising with auditory models. *Interspeech*. (poster)
- **MR Saddler**, J Feather, A Franci, JH McDermott. Hearing-impaired artificial neural networks replicate speech recognition deficits of hearing-impaired humans. *Virtual Conference on Computational Audiology*. (talk)
- **MR Saddler**, R Gonzalez, JH McDermott. Deep neural networks reveal interplay of peripheral coding and stimulus statistics in human pitch perception. *Association for Research in Otolaryngology*. (poster)
- **MR Saddler**, JH McDermott. The role of auditory nerve phase-locking in human hearing: evidence from deep neural networks. *Association for Research in Otolaryngology*. (poster)

2020

- **MR Saddler**, R Gonzalez, JH McDermott. Deep neural networks reveal interplay of peripheral coding and stimulus statistics in human pitch perception. *Advances and Perspectives in Auditory Neuroscience*. (poster)
- Medina B, **MR Saddler**, JH McDermott. Investigating deep artificial neural networks trained to do ecological tasks as a normative model for pitch perception. *Society for Advancement of Chicanos/Hispanics and Native Americans in Science*. (poster)
- Medina B, **MR Saddler**, JH McDermott. Investigating deep artificial neural networks trained to do ecological tasks as a normative model for pitch perception. *Central European Conference on Information and Intelligent Systems*. (poster)
- **MR Saddler**, J Feather, A Franci, JH McDermott. Hearing-impaired deep neural networks replicate behavioral deficits of hearing-impaired humans. *COSYNE: Computational and Systems Neuroscience*. (poster)
- **MR Saddler**, J Feather, A Franci, R Gonzalez, JH McDermott. Deep neural networks with simulated hearing impairment replicate behavioral deficits of hearing-impaired listeners. *Association for Research in Otolaryngology*. (talk)

2019

- **MR Saddler**, R Gonzalez, JH McDermott. Characteristics of human pitch perception emerge in neural networks optimized to estimate F0 from natural sounds. *Advances and Perspectives in Auditory Neuroscience*. (poster)
- JH McDermott, J Feather, A Franci, **MR Saddler**, Y Zhang, H Sarker. Next-generation hearing aids via neural network models of the auditory system. *MIT-IBM AI Horizons Colloquium*. (poster)

- **MR Saddler**, R Gonzalez, JH McDermott. Neural networks trained to estimate F0 from natural sounds replicate properties of human pitch perception. *Association for Research in Otolaryngology*. (poster)

2018

- **MR Saddler**, R Gonzalez, JH McDermott. Neural networks trained to estimate F0 from natural sounds replicate properties of human pitch perception. *MIT Quest Symposium on Robust, Interpretable Deep Learning Systems*. (Poster)
- **MR Saddler**, R Gonzalez, JH McDermott. Neural networks trained to estimate F0 from natural sounds replicate properties of human pitch perception. *Speech and Audio in the Northeast*. (Poster)

2017

- **MR Saddler**, A Bocconcelli, LS Hickmott, G Chiang, R Landea-Briones, PA Bahamonde, G Howes, LS Sayigh. Characterizing Chilean blue whale vocalizations with digital acoustic recording tags: a test of using tag accelerometers for caller identification. *Acoustical Society of America & European Acoustics Association*. (poster)

2016

- A Bocconcelli, **MR Saddler**, LS Hickmott, G Chiang, PA Bahamonde, F Caruso, LS Sayigh. Acoustic behavior of blue whales (*Balaenoptera musculus*) in the Gulf of Corcovado, Chile, recorded on DTAGs. *The 4th International Conference on the Effects of Noise on Aquatic Life*. (poster)